

Brute Force: Password Guessing (T1110.001)

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1 Attack Description in CNL

The attack description of the MITRE ATT&CK Technique “Brute Force: Password Guessing” (T1110.001)¹ written in the proposed CNL is shown below.

```
1 Tactic: TA0006 Credential_Access
2 Technique: T1110.001 Brute_Force_Password_Guessing
3
4 Background
5 Given process as AuthenticationService
6 And account as TargetAccount
7 And message as CorrectCredentials
8 And message as IncorrectCredentials
9 And session as User
10
11 Event 1
12 Given AuthenticationService is awaiting_input
13 When AuthenticationService receives IncorrectCredentials to TargetAccount
14 Then IncorrectCredentials is rejected
15 Repeated 5 times within 1 sec
16
17 Event 2
18 Given IncorrectCredentials is rejected
19 When AuthenticationService receives CorrectCredentials to TargetAccount
20 Then CorrectCredentials is accepted
21 And User is active
22
23 Completion
24 Event 2
```

2 Attack Model (Petri Net)

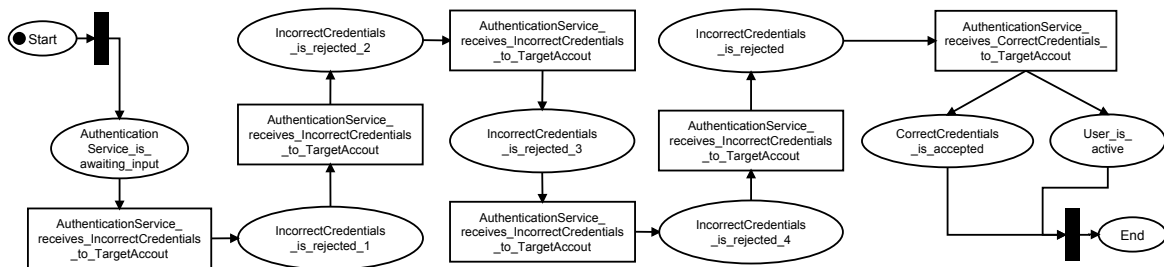


Figure 1: Brute Force: Password Guessing Petri Net

¹See <https://attack.mitre.org/techniques/T1110/001/>

3 Runtime and Memory Usage

The translation of this CNL description took 550 ms and allocated 480 KB at peak. The translation pipeline was run on a laptop with Intel Core i7-12650H, 16 GB RAM and running Windows 11.

4 Validation Criteria

The validation process included semi-structured interviews with two domain experts, in which the following validation criteria were discussed: **V1.** the technique description (in CNL) expresses the content of the ATT&CK text; **V2.** the technique description (in CNL) does not contain ambiguities; **V3.** the Technique Model captures the described attack features; **V4.** the Petri Net presents the alternative paths the attack can follow based on the described actions, pre- and post-conditions and assets involved. In the discussion section below, the domain experts are referred to as E1 and E2.

5 Discussion

5.1 V1 & V2

The CNL description for *Password Guessing* provides a clear expression of the events that constitute the attack, along with its participating assets. Additionally, the CNL description explicitly states the frequency of the login event with a specific number of repetitions (5) and a time duration in which the repetitive action occurs (1 second). This addresses directly the ambiguity found in the MITRE ATT&CK description. The assets involved in the attack are described by using only their type and name, and no properties have been declared for any of them. The CNL description stays faithful to the source, since it follows the high level of abstraction the ATT&CK description uses regarding the assets. During the interview, E1 stated that the way the attack has been described maintains the natural language, addresses the ambiguity and provides better operational information on the detection of the attack. E2 also mentioned that even though this CNL description is just a demonstration, it contains more explicit information on the system behavior under the attack than ATT&CK.

5.2 V3 & V4

The Technique Model produced by the CNL description accurately preserves the attack features as stated in the description and the Petri Net presents the system behavior under the password guessing attack. In particular, the Petri Net showcases the repetitive occurrence of the login event with multiple transitions stating each login attempt, until a successful one finishes the attack. Overall, the actions and state conditions are presented as they are described in the CNL text. The intermediate places between the repeated actions represent the same state condition, and are followed by an identifier to distinguish the different places, appropriately showing the bounded repetition of the action. While evaluating the results, E2 highlighted that they can clearly see how the attack plays out and how they can monitor such behavior to effectively detect the attack.